# THE LEGAL STATUS OF GEOSTATIONARY ORBIT: SOME REMARKS

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## *A. Introductory Remarks*

Incessant progress of the space science and technology produces new possibilities for the use of outer space and as a result the need for new legal regulations has arisen. A significiant example of this is the problem of the use of the so-called geostationary orbit.

The geostationary orbit is a circular orbit located at a distance of about 35,800 kilometers over the earth's equator. A satellite placed in this orbit turns about the polar axis of the earth in the same direction and with the same period as that of the earth's rotation. The positioning of artificial satellites in the geiostationary orbit is now of great practical importance for telecommunications. But, it is expected that this orbit will be used in the near future for several other applications.[1]

The question of utilization of the geostationary orbit is complicated by the fact that due to technical considerations, the number of artificial satellites, which may be placed in it without causing mutual interferences, is limited.

The first artificial satellite, Syncom 2, was placed in geostationary orbit by NASA on the 26th of July 1963. By July 1977 the total number of geosynchronous satellites reached about one hundred. In the opinion of several experts in space technology, no more than 180 space objects can be placed in geostationary orbit.[2]

The important task of developing an adequate system for the utilization by different States of the geostationary orbit was undertaken by the International Telecommunication Union. The matter was also discussed in detail at the World Administrative Radio Conferences at Geneva in 1971 and 1977. The Final Act of the latter, signed on 13 February 1977 by the representatives of 106 countries, contains a plan assigning positions in the geostationary orbit for broadcasting satellites and

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[1]See L. Perek, Physics, Uses and Regulation of the Geostationary Orbit, or, *Ex Facto Sequitur Lex.* Doc. IAF-SL-77-44, p. 20-22.

[2]Some other experts consider this maximum number to be greater.

frequency channels in the 11.7-12.2. GHz band to States in the ITU Region 1 (Europe and Africa) and 3 (Asia).[3]

The juridical aspects of the utilization of the geostationary orbit has been previously treated in space law literature.[4] But with the advent of claims to exclusive sovereignty by equatorial countries over segments of this orbit, it has become the focus of heated discussion.

### *B. Claims of the Equatorial Countries*

It was Colombia which during the thirtieth session of the United Nations General Assembly in October 1975 first laid claim to a segment of the geostationary orbit lying over its national territory. The representative of Colombia argued that this segment is a part of the territory of his country and that "it is not included in the conception of outer space alluded to in the Treaty on Principles Governing The Activities of States in The Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies, signed in January 1967".[5]

This stand of Colombia was again expressed during the next session of UN General Assembly in 1976. At the same thirty-first session an analogous position was taken by Ecuador[6] and Panama.[7]

On November 29, 1976 a special conference of eight equatorial countries convened in Bogota, Columbia. This conference ended on December 3, 1976 with the signing of a common Declaration setting forth in a systematic and detailed manner these nations' position on the legal status of the geostationary orbit.[8]

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[3]This Act will enter into force on January 1, 1979. For the ITU Region 2 (Americas) an interim plan was adopted for the period pending the adoption of a more detailed plan by a Regional Administrative Radio Conference to be held no later than 1982.

[4]See J. Busak, Geostationary Satellites and the Law, Telecommunication Journal (No. VIII, 1972); M. G. Marcoff, Traite de droit international public de l'espace, 64, 232, 618, n. (Fribourg, 1973); B.G. Dudakov, Pravovye problemy ispolzovanya orbit dla iskustvennych sputnikow Zemli, "Sovietskoje Gosudarstvo i Pravo" 78 (No. 4, 1973), n.; G. C. Merckel, The Direct Broadcast Satellite: The Need For Effective International Regulation, 2 Syracuse J. Int'l L. and Commerce 111 (1974).

[5]UN Doc. A/C.1/PV.2052, pp. 45-46.

[6]UN Doc. A/C.1/31/PV.10, pp. 37-38.

[7]UN Doc. A/C.1/31/PV.10, pp. 81-82.

[8]The representative of Brazil signed the Declaration as an observer. For a text of the Declaration, see EL ESPECTADOR (Columbia), December 7, 1976, p. 13A. For the English translation see ITU, Broadcasting Satellite Conference, Doc. No. 81-E (January 17, 1977). 6 J. Space L. 193.

As a starting point the signatories state that "the geostationary synchronous orbit is a physical fact linked to the reality of our planet because its existence depends exclusively on its relation to gravitational phenomena generated by the earth and that is why it must not be considered part of the outer space". This basic statement leads them to the principal point of the Declaration - a proclamation of their national sovereignty over the respective segments of the geostationary orbit as being their "natural resource". Positioning of devices in these segments shall require - according to the Declaration - "previous and expressed authorization on the part of the concerned State".[9]

The next paragraphs of the vast document try to explain the motivation for this advanced claim.

### *C. Arguments of the Equatorial Countries*

The first argument deals with the assertion that the rights of equatorial countries to treat segments of the geostationary orbit as being under their national sovereignty is founded on the fact that these segments, linked with their terrestrial territories by earth's gravitation, constitute one of the natural resources which the Resolutions of the United Nations General Assembly has recognized as belonging to States and especially to the developing countries as part of their "full and permanent sovereignty".[10]

The second argument is formulated by way of deduction from the Space Treaty of 1967. The geostationary orbit may be considered as being not a part of outer space in the meaning of this Treaty because this Treaty had not established a definition of outer space. From this lack of such a definition, the signatories of the Bogota Declaration draw a conclusion that Article II of the Treaty, forbidding any national appropriation of outer space, does not apply to the geostationary orbit.

It is characteristic and somewhat confusing that the Bogota Declaration of December 3, 1976 uses the Space Treaty of 1967 for drawing the above mentioned argumentation and contests its value by saying that it "cannot be considered as a final answer to the problem of the exploration and use of outer space". This is particularly so since its text was "elaborated when the developing countries could not count on adequate scientific advice and were thus not able to observe and evaluate the omissions,

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[9]As to the segments of the geostationary orbit over the open sea the equatorial countries declared that they consider these segments as the "common heritage of mankind" since they are beyond the national jurisdiction of states. *Ibid.*

[10]On this point, the signatories of the Declaration quote the UN General Assembly Resolution 2692/XXV on. "Permanent sovereignty over the natural resources of developing countries and expansion of domestic sources of accumulation for economic development" and the UN General Assembly Resolution 3281/XXIX on "The Charter on Economic Rights and Duties of States". *Ibid.*

contradictions and consequences of the proposals which were prepared with great ability by the industrialized powers for their own benefit''.

*D. Discussion in the UN Outer Space Legal Subcommittee*

The equatorial countries reitereated their claims to parts of the geostationary orbit at the ITU World Radio Conference held at Geneva in January-February 1977. Several States participating in this Conference made formal declarations that ''the assignment of positions in the geostationary orbit for broadcasting satellites are fully in conformity with the generally recognized principles and rules of international law''.[11]

In the United Nations Outer Space Legal Subcommittee the question was debated for the first time during its Sixteenth Session held in New York from March 14, to April 8, 1977. It was debated in connection with two agenda items: ''Elaboration of principles governing the use by States of artificial earth satellites for direct television broadcasting'' and ''Matters relating to the definition and/or delimitation of outer space and outer space activities''.

Three equatorial countries; Colombia, Ecuador and Guyana which at the time were not members of the UN Committee on the Peaceful Uses of Outer Space and of its Scientific and Technical, and Legal Subcommittees obtained at their request the right to attend the formal meetings of the latter.[12]

The most extensive statement on behalf of the equatorial countries was that of the Colombian representative Mr. E. Gaviria who presented in detail the position of the signatory States of the 1976 Bogota Declaration. He proposed to convene a special international conference for considering ''with the proper care and seriousness the definition of outer space and the special regime called for by phenomenon of the geostationary synchronous orbit''. Maintaining that the proclamation of the national sovereignty over segments of the geostationary orbit is not contrary to the provisions of the 1967 Space Treaty, the Colombian delegate contended simultaneously that this Treaty ''did not take account of the interests of the developing countries'' and sought rather to ban the use of space for military purposes than to deal appropriately with the phenomenon of telecommunications.[13]

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[11]See ITU, Broadcasting Satellite conference Doc. 266/Rev. 1/-E.

[12]UN Doc. A/AC.105/C.2/SR. 266, p. 2.

[13]UN Doc. A/AC.105/C.2/SR.77, p. 2-4. The representative of Colombia also reproached the UN Outer Space Legal Subcommittee that it served ''the interests of the highly industrialized states, harmed those of the developing countries''. This reproach produced a firm opposition on the part of Mr. B. G. Maiorski, representative of Soviet Union, who rightly stated that the Subcommittee acted in accordance with the principle of consensus and ensured that the views of all countries, both developed and developing, received an equal hearing and equal respect. *Id.* at 5.

Ambassador M. A. Albornoz from Ecuador stated that a rejection of the claims of equatorial countries would lead to "a neocolonialism of outer space". He compared these claims to national sovereignty over segments of the geostationary orbit with those concerning the admitted jurisdiction of the coastal States over the maritime economic zone.[14]

The juridical analogy with the legal status of the maritime economic zone was also invoked by the Indonesian delegate Mrs. I. M. Damanik who proposed to conclude a treaty stipulating "the granting of priority to equatorial states in the use of the geostationary orbit".[15]

The representative of Kenya Mr. J. Simani emphasized the urgent need of the formulation of a definition of outer space taking "into account the special position of equatorial countries with respect to the geostationary orbit forming part of their natural resources".[16]

The position of equatorial countries produced criticism by a number of delegations of member States of the United Nations Outer Space Legal Subcommittee. But their statements in this matter had a general and preliminary character without detailed analysis of the various juridical aspects and implications of the problem. The delegate of the Soviet Union, Mr. B. G. Maiorski, said the geostationary orbit is inseperable from outer space and the location of States did not creat any right of ownership to it or to any segment of it.[17]

The author of this paper, acting as the representative of Poland, stated that the geostationary orbit forms an integral part of outer space and "unquestionably comes under the provisions of Article II of the Treaty of January 27, 1967" and that it "could not be subject to the exclusive sovereignty of States".[18]

More extensive pronouncements were made by the delegates of the United Kingdom and United States.

Mr. A. M. Greenwood, head of the British delegation, rightly pointed out that the claims of equatorial countries were vague as to the question of whether or not these countries claim part of the geostationary orbit as a slice taken out of the rest of outer

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[14]UN Doc. A/AC.105/C.2/SR.272, p. 3.

[15] *Id.* SR.272 at 6.

[16] *Id.* SR.280 at 2.

[17] *Id.* SR.282, at 3. A more detailed presentation of the Soviet position in this matter is contained in UN Doc. A/AC.105,L.94.

[18]UN Doc. A/AC.105/C.2/SR.277, p. 2.

space or whether their claims included the whole of the space segment between the underlying country and the geostationary orbit.[19]

Mr. D. P. Stewart, United States delegate, criticized with insight the assertion that the legal admissibility of the national appropriation of the geostationary orbit is based on its dependence on earth's gravitation.[20]

It is expected that the seventeenth session of the UN Outer Space Legal Subcommittee to be held at Geneva in 1978 will have a more detailed and complete discussion of the matter.

*E. Claims of National Sovereignty over the Geostationary Orbit and International Law*

An adequate juridical evaluation of the international legal status of the geostationary orbit begins with an adequate statement of its real essence.

The signatories of the 1976 Bogota Declaration classify the geostationary orbit as "a physical fact" depending exclusively on the earth's gravity. This approach is an improper one. In reality the geostationary orbit is nothing more than one of the possible trajectories of the artificial earth satellites.

The space technology experts agree that the position of an artificial satellite in geostationary orbit is dependent on several factors such as: the launch and station-keeping propulsion, the attraction of the earth, the moon and the sun, and the solar radiation pressure.[21] Therefore the force of the earth's attraction is merely one of the elements determining the maintenance of an artificial satellite in the geostationary orbit, but it is not the only one.

It is quite nonsensical to maintain that the alleged rights of a sovereign over segments of the geostationary orbit derive from the action of the gravitational force of the terrestrial territories belonging to equatorial countries. The force of the earth's gravity derives from the mass of the whole of our planet and the sub- division demanded by the equatorial countries is unfeasible and preposterous.

The rejection of this argument of the signatory States of the 1976 Bogota Declaration also overturns the thesis that parts of the geostationary orbit constitute their "natural resources".

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[19] *Id.* SR.269, at 8.

[20] Press Release USUN-18/77 (April 7, 1977).

[21] A highly precise description of these factors determining the position of the geostationary satellites in orbit was given recently Professor Lubos Perek, Chief of the United Nations Outer Space Affairs Division, in his above cited paper.

The remaining principal plea in the position of the equatorial countries is seeking a juridical base for the admissibility of national appropriation of segments of the geostationary orbit in the absence of an outer space definition or delimitation in the text of the 1967 Outer Space Treaty. This plea is juridically pointless, too.

The fact that the 1967 Treaty failed to define the scope of the term 'outer space' is not tantamount to the impossibility of determining this scope in a general manner by way of deducing it from the Treaty's provisions, as a whole.

The main intent of the Treaty was to institute a set of international legal principles governing the activities of States in the exploration and use of outer space. This use encompasses objects launched into outer space and in particular artificial earth satellites placed in orbit around the earth. Therefore, implementation of the 1967 Treaty is possible on the assumption that its provisions concern those regions of space in which the artificial earth satellites are placed. The acceptance of an opposite assumption deprives the 1967 Treaty, and other international conventions based on it, of a reason for their existence.

An admission that Sates have a right to decide at choice which parts of space above the earth they will treat as outer space under the rule of the 1967 Treaty will be tantamount to a recognition of total arbitrariness inconsistent with the essence of international legal order.

One should conclude that the legal status of the geostationary orbit cannot be different from that of the whole outer space and in consequence any national appropriation of it is inadmissible.

# AN INTEGRATED LEGAL SYSTEM FOR SPACE +

*by*
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Currently there are 560 space shuttle operations planned through 1992.[1] If each of these operations were to carry a six-man crew, 3,300 astronauts would be launched into space before the end of this century. While the space shuttle is the only means of space transportation now being geared up for operational use, plans are being considered for other methods of low cost access to the vast regions beyond the earth's atmosphere. As cheaper transportation provides greater access to space, entry into space by commercial firms is certain to expand.[2] Some experts predict space industry revenues will reach $20 billion annually by the year 2000.[3] Products and processes already being considered are: satellites (for communication, remote sensing and solar power), crystal growth, pharmaceuticals, alloying of metals and ball bearings, electronic components, and ultrapure exotic materials that can be produced in zero gravity.[4] A recent survey shows that a sizeable selection of U.S. companies are already contemplating spaceborne activity. Boeing has conducted a study for NASA that involves converting the sun's rays to electrical power and beaming them to earth. The plan would require scores of shuttle flights to build a construction base in space and would require a five hundred man construction crew for about a year to build a solar power satellite.[5] In a more conjectural vein, Professor O'Neill of Princeton proposes a beachhead manufacturing plant in space with a large factory of workers living within a one-mile circumference of the plant.[6] It is

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+ This paper is based on a presentation by the author at the 1977 Annual Meeting of the American Astronautical Society in San Francisco.

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[1]Hearings on H.R. 2221 Before the House Subcommittee on Space Science and Applications of the Committee on Science and Technology, 94th Cong., 2d Sess., (1976) [1 1978 NASA Authorization, 394 (1976)] (hereinafter cited as 1978 NASA Authorization).

[2]1966 proposal for a "skyhook"; a cable from a satellite in geostationary orbit to earth. Payloads would be sent up the cable mechanically: Hearings Before the House Subcommittee on Space Science and Applications of the Committee on Science and Technology, 94th Cong., 1st Sess. (1975) (*Future Space Program 1975*, 198 (1975).

[3] *"The Shuttle Opens the Space Frontier to U.S. Industry," Bus. Week,* August 22, 1977, at 49.

[4]The most important system being considered by NASA is the Space Construction Base, *see* 1978 Authorization at 399. This space station would serve for a ten year study of space industrialization and satellite power generation. *Id.* Also being studied are: large structures in orbit via the shuttle and development of Heavy Lift Launch Vehicles. *Id.* at 400, 415.

[5] *Akron Beacon J.*, February 1, 1978, at 2.

[6]. *New Scientist,* June 23, 1977, at 720.